

Heat Transfer, Exchangers and Where Heat Goes

How heat actually moves, the one component that carries it out of the fire, and the limit on how far you inspect it.

Conduction	Heat moving through a solid material. The best example on the exam is heat flowing from the inside to the outside of a radiator.
Convection	Heat carried by a moving fluid — air or water.
Radiation	Heat crossing space directly, warming objects rather than the air between.

YOU HAVE MISSED THIS 3 TIMES

Which of the following is the best example of conduction as it relates to heat flow?

Heat flowing from the inside to the outside of a radiator.

The radiator is a deliberately awkward example, because a radiator mostly warms a room by convection. The question is not about the room — it is about the heat crossing the metal wall of the radiator itself, which is conduction through a solid.

The heat exchanger

Heat exchanger	The part of the heating system that transfers heat to air or water. It keeps combustion gases on one side and the air you breathe on the other.
Defective heat exchanger	Indicated by rust and mineral deposits on the furnace plenum.
Cause of failure	Metal fatigue and rust. It expands and contracts every cycle for twenty years.
Plenum	The large metal chamber inside the heating system where hot air builds up pressure before being forced out to the heat registers.

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A large metal conduit box located inside the heating system where hot air builds up pressure and is then forced out to the heat registers is a:

Plenum.

YOU HAVE MISSED THIS

When inspecting the heating system, an inspector is not required to:

Fully inspect the heat exchanger.

A full inspection would mean dismantling the furnace. You look at what is visible and report the indications — rust, deposits, flame disturbance — and recommend evaluation. A cracked heat exchanger leaks combustion products into the supply air, so the stakes are high and the honest answer is to refer it.

Where the heat should come out

YOU HAVE MISSED THIS 2 TIMES

The most efficient location for heat supply in a room is on a/an:
Exterior wall near a window or door.

Put the heat where the cold is. An exterior wall, and especially a window, is where the room loses heat and where cold air falls down the glass. Supplying there meets the draft at its source instead of letting it cross the room.

YOU HAVE MISSED THIS

The most important maintenance component of an electric furnace is:
The air filter.

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Home inspectors inspect solar heating systems only when:
They are qualified to do so and make that election.

Say it back

Cover the page. Answer each one out loud.

1. Conduction, convection, radiation — define each.
2. Why is the radiator the example of conduction, not convection?
3. What does a heat exchanger do, and what indicates a defective one?
4. What causes heat exchangers to fail?
5. What is a plenum?
6. How far are you required to inspect the heat exchanger?
7. Where is the most efficient place to supply heat to a room, and why?
8. When does an inspector inspect a solar heating system?